1 UART TX

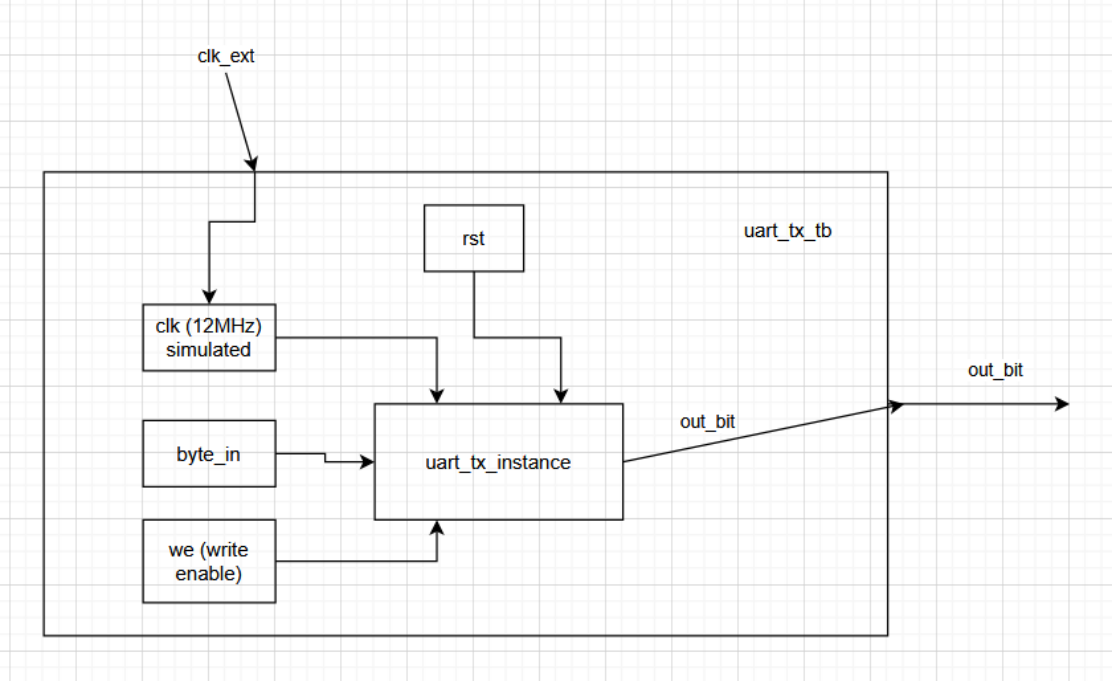

Abbildung 1: UART TX Blockdiagramm

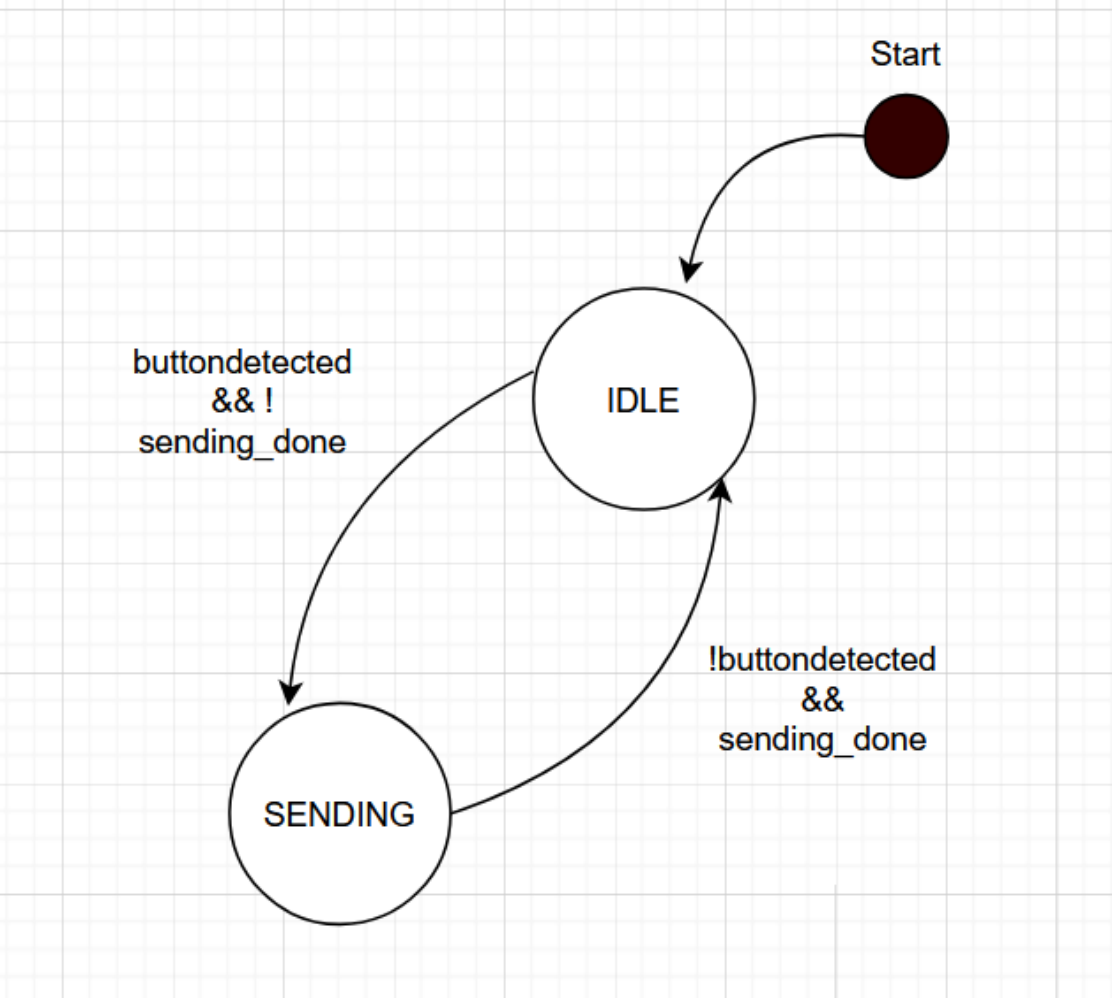

Abbildung 2: UART TX Statusmaschine

```
library ieee;
use IEEE.std_logic_1164.all;
use IEEE.numeric_std.all;

entity uart_tx is
        port(
            clk : in std_logic;
            byte_in : in std_logic_vector(7 downto 0);
            we : in std_logic;
            out_bit : out std_logic;
            rst : in std_logic
        );
    end entity;


architecture uart_tx_a of uart_tx is
    type state_type is (IDLE, SENDING);
    signal state : state_type := IDLE;
    signal symbol_cnt : integer := 0;
    constant baud_rate : integer := 9600;
    constant clk_speed : integer := 12000000;
    constant bit_period : integer := clk_speed / baud_rate;
    signal sending_done : std_logic := '0';
    signal button_detected :std_logic := '0';

begin
    process(clk) is
    variable cnt_period : integer := 0;
    variable start_bit_done : std_logic := '0';
    variable char_bits_done : std_logic := '0';
        begin
            if rising_edge(clk) then

                if rst = '1' then
                    state <= IDLE;
                    out_bit <= '1';
                else

                    if button_detected = '1' then
                        state <= SENDING;
                        button_detected <= '0';
                    elsif sending_done = '1' then
                        state <= IDLE;
                        sending_done <= '0';
                    end if;

                    case state is
                        when IDLE =>
                        if we = '1' then
                            button_detected <= '1';
```

```
51                          else
52                              out_bit <= '1';
53                          end if;
54                          when SENDING =>
55                              if start_bit_done = '0' then
56                                  out_bit <= '0';
57                                  if cnt_period < bit_period then
58                                      cnt_period := cnt_period + 1;
59                                  else
60                                      cnt_period := 0;
61                                      start_bit_done := '1';
62                                  end if;
63                              elsif char_bits_done = '1' then
64                                  out_bit <= '1';
65                                  if cnt_period < bit_period then
66                                      cnt_period := cnt_period + 1;
67                                  else
68                                      cnt_period := 0;
69                                      char_bits_done := '0';
70                                      start_bit_done := '0';
71                                      sending_done <= '1';
72                                  end if;
73                              else
74                                  if symbol_cnt < 8 then
75
76                                      out_bit <= byte_in(symbol_cnt);
77                                      cnt_period := cnt_period + 1;
78
79                                      if cnt_period > (bit_period - 1 ) then
80                                          symbol_cnt <= symbol_cnt + 1;
81                                          cnt_period := 0;
82                                      end if;
83                                  else
84                                      symbol_cnt <= 0;
85                                      char_bits_done := '1';
86                                  end if;
87                              end if;
88
89
90                      end case;
91
92                  end if;
93
94              end if;
95      end process;
96  end architecture uart_tx_a;
```

```vhdl
library ieee;
use IEEE.std_logic_1164.all;
use IEEE.numeric_std.all;

entity uart_tx_tb is
end entity;


architecture uart_tx_tb_a of uart_tx_tb is
    signal clk : std_logic := '0';
    signal byte_in : std_logic_vector(7 downto 0) := (others => '0');
    signal we : std_logic := '0';
    signal rst : std_logic := '0';
    signal out_bit :std_logic := '0';

begin
        uart_tx_inst : entity work.uart_tx
        port map(
            clk=> clk,
            byte_in => byte_in,
            we => we,
            rst => rst,
            out_bit => out_bit
        );

    clk <= not clk after 83.33 ns / 2;

    process
    begin
        -- Reset
        rst <= '1';
        wait for 100 ns;
        rst <= '0';

        -- Check IDLE
        wait for 50 ns;
        assert out_bit = '1' report "ERROR: out_bit not idle high after reset"
        ↪  severity error;

        -- 1
        byte_in <= "11010001";
        wait for 100 ns;
        we <= '1';
        wait for 100 ns;
        we <= '0';

        wait for 50 us;
        assert out_bit = '0'
            report "ERROR: Startbit not detected (Test 1)"
            severity warning;
```

```
50
51          wait for 2 ms;
52          assert out_bit = '1'
53              report "ERROR: Line not back to idle (Test 1)"
54              severity error;
55
56          -- 2
57          byte_in <= "10101010";
58          wait for 100 ns;
59          we <= '1';
60          wait for 300 ns;
61          we <= '0';
62
63          wait for 50 us;
64          assert out_bit = '0'
65              report "ERROR: Startbit not detected (Test 2)"
66              severity warning;
67
68          wait for 2 ms;
69
70          -- 3
71          byte_in <= "11110000";
72          wait for 100 ns;
73          we <= '1';
74          wait for 300 ns;
75          we <= '0';
76
77          wait for 50 us;
78          assert out_bit = '0'
79              report "ERROR: Startbit not detected (Test 3)"
80              severity warning;
81
82          wait for 2 ms;
83
84          -- 4
85          byte_in <= "00001111";
86          wait for 100 ns;
87          we <= '1';
88          wait for 300 ns;
89          we <= '0';
90
91          wait for 50 us;
92          assert out_bit = '0'
93              report "ERROR: Startbit not detected (Test 4)"
94              severity warning;
95
96          wait for 2 ms;
97
98          -- 5
99          byte_in <= "00110011";
```

```
100         wait for 100 ns;
101         we <= '1';
102         wait for 300 ns;
103         we <= '0';
104 
105         wait for 50 us;
106         assert out_bit = '0'
107             report "ERROR: Startbit not detected (Test 5)"
108             severity warning;
109 
110         wait for 2 ms;
111 
112         -- Ende
113         assert false report "Simulation finished successfully" severity note;
114         wait;
115 
116 
117     end process;
118 end architecture uart_tx_tb_a;
```